Natural Language Processing project

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1 Introduction

This project evaluates how well different large language models (LLMs) process, analyze and extract information from board-game rule texts. To challenge large language models to interpret the formal logic of real board games, they are tested on tasks including concise explanation generation, to evaluate the ability to extract fundamental concepts of the game and adapt them to best suit the targeted audience; error detection, to test the ability to extract the underlying logic and identify missing or contradictory information; estimation of game properties such as complexity, optimal player count, and mechanics.

2 Research question and methodology

In this project the following suggestions have been explored:

- Ask the model to explain the game to different kind of audiences.
- Feed the model a version of a game with missing or intentionally flawed rules and ask it to identify the issue and suggest a correction.
- Give the model a rulebook and ask it to classify the mechanics, evaluate the complexity, suggests the perfect number of players and estimate the duration. These model predictions can be compared with the data provided by BoardGameGeek (BGG).

Experiments have been performed using three different models: *LLaMA*, *Gemma* and *Qwen*, to compare results and highligth difference and similarity in results across different LLMs.

2.1 Data

Rules from 5 different board games have been collected from the producer website in pdf format. Games are a subset of the BoardGameGeek Hall of Fame and are displayed in table 1.

Table 1 List of board games used in this project. Board games denoted with an asterisk are the ones used only to generate explanations.

Game	BGG id
7 Wonders *	316377
Catan	13
Dominion	209418
Power Grid	2651
Ticket to Ride	9209

The first preprocessing step has been to extract the text using `pdftotext`. After this, the small sample size allowed a manual inspection of the results to fix formatting errors, rewrite unreadable sections and, in case of image dependent paragraphs, rephrase or remove them.

2.2 Explanation

Each model was first prompted to explain the game in, conversational terms to a child to evaluate the ability to extract fundamental concepts of the game and adapt them to best suit the targeted audience. Ages 7, 11, and 16 were chosen to cover the usual recommended range for board games and to be different enough to capture meaningful differences in language comprehension.

The models' outputs were evaluated on three different criteria:

- **readability**
- **completeness**
- **conciseness**

2.2.1 Readability

To measure how well the generated text was suitable for the selected audience, three of the most common readability measure (SMOG, Flesch-Kincaid, Dale-Chall) have been applied on the output and the resulting grade was compared with the requested one.

It is worth noting that SMOG and Flesch-Kincaid evaluate only polysillabic words and phrase length, while Dale-Chall also uses a list of common words to asses difficulty. As such none of them is able to grasp the complexity of the text from a logic stand point, meaning this score is only partially estimating the problem. However more complex models were beyond the scope of this project.

2.2.2 Completeness

To evaluate completeness it was decided to assign a score to every rule described in the rule-book, sum the score of all the rules present in the explanation and divide it for the maximum score achievable.

$$\frac{\sum_{r \in R} \mathbb{I}_E(r) \cdot v(r)}{\sum_{r \in R} v(r)}$$

where R, E represent respectively the set of rules appearing in the original text and in the explanation and $v(r)$ is the value function of the rule r. The problem of finding a good candidate for v and define correctly R and E is not trivial.

Luckily previous work from [1], can be used to tackle both problems at once. Using a slightly revised version of their method it was possible not only to extract from the text a comprehensive rule-set, but also, exploiting the hierarchical structure of the output, to assign meaningful weights to each rule.

To confirm the presence of a rule inside the explanation, instead, the rule was first embedded and then, using a sliding window, compared with the output's embedding. A rule was considered present if the cosine distance between its embedding and the one of any window in the text was higher than a threshold. The usage of a sliding window is motivated by the size difference between the rule and the explanation text, since the latter contains also other content not related to the subject that would lower the overall score leading to possible detection problems. Empirically the threshold value was set to 0.5, the window size to 10 and the stride to 5. The model of choice for the sentence embedding was *all-MiniLM-L6-v2*[2] for its good tradeoff between quality and inference speed.

2.2.3 Conciseness

To evaluate conciseness the output length was simply compared to the original rule-book length. A high compression rate (assuming a good evaluation on the other two criteria) generally leads to an easier to assimilate explanation. By contrast, repeating word by word the original document, could probably result in good scores on the two other metrics but would not be an helpful explanation.

2.3 Error Detection

For each game, models were fed the original rule-book and a set of four intentionally flawed copies, each differing from the original rule-book for one single error. The ratio was to have different types of edit increasingly difficult to spot, and check for false positives using the unaltered rule-book.

Here is the guideline followed for the 5 different level of edit:

0. **original:** the unaltered rulebook
1. **missing:** an entire paragraph of the rulebook describing some core mechanic is missing
2. **unsolvable:** a rule directly contradicts another one
3. **incoherent:** a combination of rules hardlocks the game
4. **game-breaking:** a coherent but obviously unbalanced mechanic

For each of these levels, 5 different outputs were generated with the same prompt to have a more statistically relevant measure. To test the results given from the each model another LLM was prompted to compare the ground truth and the output. For this task the *Qwen* has been chosen, as this was slightly faster than the other two and more capable of reliably producing well formatted text.

2.4 Parameter Estimation

For this experiment each model was prompted to estimate the following parameters for each game:

- A list of the game mechanics
- Game complexity
- Best number of players
- Duration

For this task, 5 different prompts have been made. The first four, one for each parameter, supply the model a small chain of thought to follow and the formatting for the last line of the output used for the final output. Finally one last prompt was used to test both the ability of the model to estimate all parameters using only one prompt. For this last prompt the model was instructed to output a JSON file following a predefined structure. For each combination of prompt, game and model 10 different output were generated in order to obtain more statistically relevant data. Sometimes, even with restriction in both prompt and response format, output does not include the last line or is not a well formatted JSON. For this reason a check is performed after each output generation and in case of a malformed output up to 10 retry are attempted.

To evaluate each model, outputs were then compared with BGG's values. In case of mechanics estimation since a full BGG's mechanics list was provided only in the single parameter prompt, LLM's outputs are often non-existent but believable mechanics. In this case two distinct evaluation method have been considered: the classic one where one mechanic is considered correct if it appears in the ground truth, and a less rigorous one, where the mechanic is considered correct if the name is different but if expresses the same concept. For estimating this last property both the predicted and real mechanic are embedded and are considered similar if the the cosine distance is ≤ 0.4

3 Experimental results

3.1 Explanation

As shown in the table 2, even though there is a small metrics' increase with the age progression, the variation is really small across all models on any metric, meaning that the model output is not really adjusted based on the target age. Moreover, the score are quite high, signaling the inability to correctly simplify the lexicon. This last problem is especially evident when using a readability metric that also takes into account common words like the Dale-Chall score.

It is important to note that while Flesch-Kincaid and SMOG scores can be compared directly with the target grade (column 'Expected'), Dale-Chall needs a conversion table to interpret the score (column 'Dale-Chall') as a grade (column 'Dale-Chall grade').

Model	Age	Expected	Flesch–Kincaid	SMOG	Dale-Chall grade	Dale–Chall
LLaMA	7	2	4.68	7.72	10-12	8.25
	11	6	5.76	8.47	10-12	8.46
	16	11	6.31	9.12	12	8.99
Qwen	7	2	4.41	7.09	10-12	8.13
	11	6	4.70	7.58	10-12	8.56
	16	11	5.35	8.25	>12	9.31
Gemma	7	2	3.68	6.97	9-10	7.70
	11	6	4.41	7.81	10	8.01
	16	11	4.81	8.08	10-12	8.36

Table 2 Readability metrics for LLaMA, Qwen, and Gemma across three age groups

Table 3 shows both the mean rule coverage and relative size for the explanations. As expected, there is a clear tradeoff between the two, this especially penalize rule coverage for rule-intensive games like 'Power Grid'. Note that while the relative size is really low for 'Dominion', in reality the game is less complex than this data might suggest, since most of the text length is composed by card descriptions, which does not contribute much to the overall complexity. This is a sign that in this case models are able to correctly extract the most important informations from the text.

Overall LLaMA seem to be the model with the best ratio between coverage and size.

3.2 Error detection

This is the experiment where the limitation of the smaller models is most evident. In fact no model was able to spot flaws in the rule text except for *GPT-oss*. Unexpectedly the first level of edit was particularly difficult to spot even though in some cases almost as much as one third of the rulebook was removed. Performance on unmodified texts is instead more predictable. This aligns with the tendency of LLMs to identify flaws even when they are not actually present and can be probably improved with better prompting. *Qwen* was the only model that was able to correctly label the unmodified text even outperforming the much bigger *GPT-oss*.

3.3 Estimation

While predicted complexity is more or less proportional to the real value, models tends to overestimate the complexity of games, especially simpler ones. It is also possible to note from table 4 that surprisingly the dedicated prompt is linked to slightly worse

Game	Model	Rule coverage ↑	Relative Size ↓
7 Wonders	gemma	0.09	10.13
	llama	0.10	9.90
	qwen	0.07	10.68
Catan	gemma	0.17	11.86
	llama	0.18	12.28
	qwen	0.20	14.55
Dominion	gemma	0.39	6.61
	llama	0.36	6.75
	qwen	0.40	9.91
Power Grid	gemma	0.11	5.68
	llama	0.11	6.22
	qwen	0.10	5.50
Ticket to Ride	gemma	0.76	16.38
	llama	0.91	19.59
	qwen	0.90	23.32

Table 3 Explanation completion and relative size

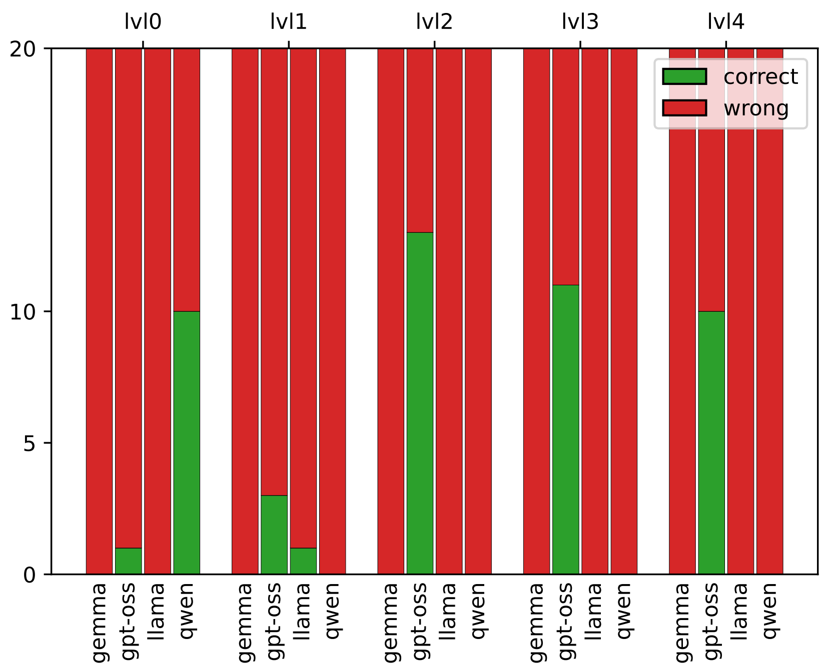

Fig. 1 Correctly detected errors for each level and model

Model	Mean abs error Complexity	Optimal players	Duration
llama	1.69	0.42	29.25
llama_all	1.23	1.05	33.00
qwen	1.22	0.30	32.40
qwen_all	1.06	0.58	24.38
gemma	1.09	0.33	20.38
gemma_all	1.05	0.48	35.88

Table 4 Mean absolute error of the parameter estimation for each model and prompt

performances for this parameter, in contrast to the behavior of the other two. The right optimal number of players was predicted fairly consistently especially with the dedicated prompt. It is important to note that this parameter has a lower range of acceptable values with respect to the other two, nonetheless, all models correctly estimated this range (not explicitly stated in the rules) and almost all prediction were inside of it.

On the contrary, duration estimation was extremely unreliable with high mean error and prediction variance.

Regarding Mechanic estimation, models were often unable to output valid mechanics even with the dedicated prompt, however also accounting for similar concepts yielded a sensible increase in performance. With loose matching the best performing model, **Gemma**, was able to correctly predict half of the correct mechanics, even though less than one third of its prediction were correct.

	Exact match		Loose match	
	mean precision	mean recall	mean precision	mean recall
gemma	0.144770	0.276071	0.256993	0.528810
llama	0.105373	0.119722	0.288261	0.351944
qwen	0.066857	0.059048	0.222447	0.230952

Table 5 Model precision and recall (exact and loose).

4 Concluding remarks

Overall the size of this model was a clear limiting factor. The models proved to be relatively good at estimating some game parameters like best player number and mechanics, as well as extract rules from games. On the other hand, they were unable to detect even obvious rule fallacies, where a larger model did not have the same problems. It would be interesting to test larger or fine-tuned models and compare results.

References

[1] Li, D., Zhang, S., Sohn, S.S., Hu, K., Usman, M., Kapadia, M.: Cardiverse: Harnessing LLMs for Novel Card Game Prototyping (2025). https://arxiv.org/abs/2502.07128

[2] Reimers, N., Gurevych, I.: Sentence-bert: Sentence embeddings using siamese bert-networks. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, ??? (2019). https://arxiv.org/abs/1908.10084